

Algebra 1: Second Semester Final Review

Chapters 6, 7, and Rad

1. When multiplying same bases what do you do to the exponents? *Add*
2. When dividing same bases what do you do to the exponents? *Subtract*
3. When you have a power raised to another power what do you do to the exponents? *multiply*
4. Anything raised to the zero power is equal to what? *one*
5. How do you undo a negative power?

Simplify the expression. Write your answer using only positive exponents.

6. $\frac{2^{-3}x^0}{y^{-4}}$ $\frac{y^4}{2^3} = \frac{y^4}{8}$

7. $(-4x^2)^3 (-4)^3 x^6$
 $-64x^6$

8. $5^{-6} \cdot 5^3$
 $5^{-3} = \frac{1}{5^3} = \frac{1}{125}$

9. 3^0
 1

10. $6xz^{-2} \cdot x^2z$
 $6x^3z^{-1} = \frac{6x^3}{z}$

11. $12c^{-7}d^6$
 $\frac{12d^6}{c^7}$

Solve the equation.

12. $3^{x-2} = 9^{2x+3}$
 $3^{x-2} = (3^2)^{2x+3}$
 $x-2 = 4x+6$
 $-8 = 3x$
 $-\frac{8}{3} = x$

13. $25^{2x-3} = 25^{x+1}$
 $2x-3 = x+1$
 $x = 4$

14. Define exponential function:

An equation where your variable is the exponent.

15. Does the following table represent an exponential function?

a)

-2	2
-1	4
0	8
1	16

$$\frac{8}{4} = 2$$

$$\frac{4}{2} = 2$$

yes

b)

-2	2
-1	4
0	6
1	8

NO

16. You deposit \$500 in a savings account that earns 7% interest compounded annually. $A = P \left(1 + \frac{r}{n}\right)^{nt}$

- A) Write the function that represents the balance after t years.
B) What is the balance after 2 years?

$$A = 500 \left(1 + \frac{.07}{1}\right)^{1(t)}$$

$$A = 500(1.07)^2$$

$$A = \$572.45$$

17. You buy a used car for \$6599. Its value decreases by 12% every year.

- A) Write the function that represents the value y (in dollars) of the car after t years.

- B) What is the value of the car after 20 years?

$$A = 6599 \left(1 - \frac{.12}{1}\right)^t$$

$$A = 6599(1-.12)^{20} = \$511.84$$

18. Write the function in $y = ab^x$ form that matches the situation: A population of 520,450 seagulls is increasing 6% yearly.

$$y = 520450(1.06)^x$$

19. Determine if the function is an exponential growth or decay or neither. $f(t) = 5(.13)^t$

decay

20. Determine if the function is an exponential growth or decay or neither. $f(t) = 3(1.5)^t$

growth

21. Write the radical expression in exponential form: $\sqrt[4]{8}$

$$8^{\frac{1}{4}}$$

22. Simplify each: a) 2^{-1}

$$\frac{1}{2}$$

b) $4^{\frac{1}{2}}$

$$(\sqrt{4})^1 = 2$$

c) 3^{-3}

$$\frac{1}{3^3} = \frac{1}{27}$$

Chapter 7

Write the polynomial in standard form. Identify the degree of the polynomial. Then classify the polynomial by the number of terms.

23. $5 + x^2 - 7x$

$$x^2 - 7x + 5$$

Quadratic
trinomial

24. $2x - 4x^5 + 3x^9 - 1 + x^2$

$$3x^9 - 4x^5 + x^2 + 2x - 1$$

degree: 9
polynomial

Find the sum or difference.

25. $(-5x^2 - 2) + (3x^2 + 7)$

$$-2x^2 + 5$$

26. $(x^2 + 4 - 2y^3) - (3y^2 - 7y + 4y^3)$

$$-4y^3 - 2y^2 + 7y + 4$$

$$27. (-x + 3) + (-11x^2 - 8 + 12x)$$

$$\boxed{-11x^2 + 11x - 5}$$

$$28. (2b^3 - 5b) - (7b + 3b^2)$$

$$\boxed{2b^3 - 3b^2 - 12b}$$

Find the product.

$$29. 2(x - 1)(2x + 2)$$

$$(2x - 2)(2x + 2)$$

$$4x^2 + 4x - 4x - 4$$

$$\boxed{4x^2 - 4}$$

$$30. (a + b)^2 (a + b)$$

$$a^2 + ab + ab + b^2$$

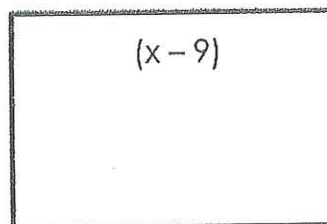
$$\boxed{a^2 + 2ab + b^2}$$

31. Find the area of the rectangle.

$$(x - 9)(x + 9)$$

$$x^2 + 9x - 9x - 81$$

$$\boxed{x^2 - 81}$$



Simplify:

$$32. \sqrt{27}$$

$$\sqrt{9} \sqrt{3}$$

$$\boxed{3\sqrt{3}}$$

$$33. \sqrt{24}$$

$$\sqrt{4} \sqrt{6}$$

$$\boxed{2\sqrt{6}}$$

$$34. \sqrt{16x^4}$$

$$\boxed{4x^2}$$

$$35. \frac{5}{\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}}$$

$$\frac{5\sqrt{5}}{\sqrt{25}} = \frac{5\sqrt{5}}{5} = \boxed{\sqrt{5}}$$

FACTOR

$$36. 4s^2 - 1$$

$$\boxed{(2s - 1)(2s + 1)}$$

$$37. x^2 - 3x - 28$$

$$\boxed{(x - 7)(x + 4)}$$

$$38. 9x^2 - 100$$

$$\boxed{(3x - 10)(3x + 10)}$$

$$39. x^2 - 13x + 40$$

$$\boxed{(x - 5)(x - 8)}$$

$$40. a^2 + 10a + 16$$

$$\boxed{(a + 8)(a + 2)}$$

$$41. (2x^3 - 4x^2)(-x + 2)$$

$$2x^2(x - 2) - 1(x - 2)$$

$$\boxed{(2x^2 - 1)(x - 2)}$$

